

MATTHEW D. HALL

EDUCATION

The University of Sheffield

PhD – Automatic Control & Systems Engineering.

OCT 2017 – PRESENT, SHEFFIELD

Lancaster University

MEng 1st class honours – Electrical & Electronic Engineering.

OCT 2012 – JUL 2016, LANCASTER

RESEARCH

Natural Robotics Lab & Sheffield Robotics

Postgraduate Research Student

PROJECT: Self-Reconfiguration of Modular Robots via Active Subtraction.

SUPERVISORS: Dr Roderich Gross (Natural Robotics Lab).

Dr Shuhei Miyashita (Microrobotics Lab).

INTERESTS: Modular Robotics, Reconfigurable Robots, Distributed Control, Industry Applications, Space Exploration.

PUBLICATIONS

M. D. Hall, A. Özdemir, R. Groß, “**Self-Reconfiguration in Two-Dimensions via Active Subtraction with Modular Robots**”, in *Proc. Robotics: Science and Systems XVI (RSS)*, RSS Foundation, 2020.

A. Özdemir, M. Gauci, A. Kolling, M. D. Hall, R. Groß, “**Spatial Coverage Without Computation**”, in *Proc. 2019 International Conference on Robotics and Automation (ICRA)*, IEEE, 2019, pp. 9674–9680.

ORAL PRESENTATIONS

2021: Departmental Postgraduate Research Symposium – *Virtual*.

2020: Int’l. Conf. on Robotics: Science and Systems (RSS) – *Virtual*.

2019: Self-Configuring Modular Robotics for Earth and Space – *York, UK*.

Int’l. Conf. on Robotics and Automation (ICRA) – *Montreal, Canada*.

UK & I Conf. on Transformative Robotics for Industries – *London, UK*.

ACSE Faculty Researcher Symposium – *Sheffield, UK*.

AWARDS & SCHOLARSHIPS

The University of Sheffield and EPSRC Doctoral Training Partnership National Productivity Scholarship

For course fees and maintenance - 2017-2021

ACADEMIC EXPERIENCE

Peer Reviewer for International Conferences

Autonomous Agents and Multiagent Systems (AAMAS), Advanced Networks and Telecommunication Systems (ANTS), Distributed Autonomous Robotics Systems (DARS), Intelligent Robots and Systems (IROS), Multi-Robot Systems (MRS), Robotics: Science and Systems (RSS), Towards Autonomous Robotic Systems (TAROS).

Graduate Teaching Assistant for Bachelor's and Master's modules

Robotics and Autonomous Systems, Mobile Robotics and Autonomous Vehicles, Robotic Systems, Digital and Embedded Systems, Foundations of Robotics.

PROFESSIONAL EXPERIENCE

Tourism Advisor – Use-it Brussels

JAN 2017 – AUG 2017

Part-time work within a sustainable tourism project, offering tourist information and running tours of the lesser-known parts of Brussels. A great opportunity to thoroughly learn about the city and its culture, as well as improving my interpersonal skills.

Student Ambassador – Lancaster University

OCT 2014 – JUN 2016

Part-time employment assisting with university events and the regular running of tours around the university campus. These responsibilities required excellent communication and organisational skills, and a personable yet professional demeanour.

SKILLS & INTERESTS

PROGRAMMING: Proficient in Python, C/C++, LaTeX, Git, Shell Scripting.
Working knowledge of MATLAB, Java, HTML, CSS and VHDL.
Released a small number of customisation apps for Android.

MUSIC: Bassist and guitarist. Experience in Ableton music production.
Regular guest radio DJ for “Clear A Space” on GagaRadio.

FILM: Keen interest in film. Written, shot and edited amateur films with family and friends.

OTHER: Food and cooking, games (board or video), craft beer.